# Binary Consensus Does Not Improve Circuit Discovery

Anonymous Authors
*Anonymous Institution*

**Abstract**

Mechanistic interpretability increasingly relies on automated circuit discovery methods, yet different methods often identify divergent circuits for the same task. A natural response is to take the consensus — edges agreed upon by multiple methods — as a higher-fidelity circuit. We systematically evaluate this binary consensus strategy (CSS) across 4 tasks on GPT-2 Small and 1 task on Qwen-0.5B, using three discovery methods (EAP-IG, Edge Pruning, ACDC). Across all six task×model configurations tested, CSS provides *no statistically significant advantage* over the best individual method ($0/6$ significant positive $\Delta$CPR at $\alpha$=0.05). The sole apparent success (IOI on GPT-2) dissolves under size-controlled baselines: top-$k$ edges from any single method match or exceed CSS performance when circuit size is held constant. We identify two mechanistic failure modes: (1) ACDC's `isinf`-based binarization collapses consensus to a single method's output, and (2) component-level scoring granularity on models with >100K edges produces near-zero inter-method overlap regardless of method independence. These findings challenge the implicit assumption that multi-method agreement validates circuit quality, and suggest that the consensus-as-validation paradigm is fundamentally flawed for current circuit discovery methods.

## 1 Introduction

Automated circuit discovery aims to identify the minimal subnetwork (circuit) within a neural network that is responsible for a specific behavior [Conmy et al., 2023, Wang et al., 2023]. As multiple discovery methods have emerged — including gradient-based attribution (EAP-IG; Syed et al., 2023), edge pruning (EP; Li et al., 2024), and ACDC [Conmy et al., 2023] — practitioners face a fundamental question: when methods disagree, which circuit should we trust?

An intuitive answer is *consensus*: edges identified by multiple independent methods should be more reliable than those found by any single method alone. This reasoning mirrors ensemble methods in machine learning and feature selection stability analysis [Nogueira and Brown, 2016, Kuncheva and Whitaker, 2003]. In circuit discovery, binary consensus takes the intersection (or super-majority) of binarized circuit outputs from $K$ methods.

We call this approach the **Circuit Selection by Consensus** (CSS) framework and test it systematically. Specifically, we define CSS($\tau$) as the set of edges receiving agreement from at least $\lceil \tau \cdot K \rceil$ methods, where $\tau \in [0, 1]$ controls the consensus threshold. At $\tau$=1.0 (unanimous agreement), CSS selects only edges found by all $K$ methods.

Our results are unequivocally negative. Across six task×model configurations spanning four tasks (Indirect Object Identification, Greater-Than Arithmetic, Subject-Verb Agreement, Country-Capital Factual Recall) and two models (GPT-2 Small with 32K edges, Qwen-0.5B with 179K edges), CSS at $\tau$=1.0 *never* provides a statistically significant advantage over the best individual method. In three configurations, CSS actively *hurts* performance (significantly negative $\Delta$CPR).

We identify two distinct mechanistic failure modes that explain this universal failure:

1. **Binarization pathology** (§5.3): ACDC's `isinf`-based edge scoring produces an extremely sparse binary mask (7–14 edges out of 32K), causing CSS($\tau$=1.0) to degenerate to the ACDC output alone. Consensus becomes a single-method circuit relabeled as "multi-method agreement."

2. **Component-level scoring collapse** (§5.7): On Qwen-0.5B, both IFR and ACDC perform node-level or block-level scoring rather than edge-level discrimination. With 179K candidate edges, this produces layer-concentrated selections with zero inter-method overlap at the edge level, making consensus vacuous.

These failure modes are not implementation bugs — they reflect structural properties of existing circuit discovery methods that make binary consensus fundamentally unsuited as a validation strategy. Our contribution is threefold: (1) the first systematic negative evaluation of consensus in circuit discovery, (2) mechanistic explanations for *why* consensus fails, and (3) evidence that the community's implicit assumption — that multi-method agreement signals circuit quality — is not supported by current methods.

## 2 Related Work

**Circuit discovery methods.** Automated circuit discovery has progressed rapidly. ACDC [Conmy et al., 2023] uses activation patching with iterative pruning. EAP-IG [Syed et al., 2023] applies integrated gradients to edge attributions. Edge Pruning [Li et al., 2024] optimizes continuous edge masks via gradient descent. More recently, IFR extends activation patching to block-level interventions, and methods like PIE [Zhang et al., 2026] propose iterative pruning ensembles. Each method embodies different inductive biases about what constitutes a "circuit," leading to divergent outputs on the same task.

**Multi-method consensus.** The idea that agreement among methods signals reliability has roots in feature selection stability [Nogueira and Brown, 2016] and ensemble diversity theory [Kuncheva and Whitaker, 2003]. Mondorf et al. [2025] propose cross-method score averaging as an alternative to binary consensus. CIRCUS [Parekh et al., 2026] explores within-method threshold diversity. Our work differs by providing the first systematic *negative* evaluation of binary consensus in circuit discovery, with mechanistic explanations for its failure.

**The consensus assumption in practice.** Despite the absence of formal evaluation, multi-method agreement is widely used as informal validation in interpretability research. Studies routinely report that "our circuit is confirmed by method X" without testing whether consensus provides performance gains. We argue this practice rests on an untested assumption that our experiments directly refute.

## 3 Method: Circuit Selection by Consensus

**Notation.** Let $\mathcal{M} = \{m_1, \ldots, m_K\}$ be a set of $K$ circuit discovery methods. Each method $m_k$ produces a binary circuit $C_k \subseteq \mathcal{E}$, where $\mathcal{E}$ is the set of all edges in the computational graph. We define:

$$\text{CSS}(\tau) = \left\{ e \in \mathcal{E} \;\middle|\; \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}[e \in C_k] \geq \tau \right\} \tag{1}$$

At $\tau = 1.0$, $\text{CSS}(1.0) = \bigcap_{k=1}^{K} C_k$ (unanimous consensus). At $\tau = 2/3$ with $K = 3$, an edge must appear in at least 2 of 3 methods.

**Evaluation metric: Circuit Performance Recovery (CPR).** We evaluate circuits using CPR, defined as:

$$\text{CPR}(C) = 1 - \frac{L_{\text{ablated}}(C) - L_{\text{full}}}{L_{\text{random}} - L_{\text{full}}} \tag{2}$$

where $L_{\text{ablated}}(C)$ is the loss when all edges *not* in circuit $C$ are ablated (set to zero or mean activation), $L_{\text{full}}$ is the unablated model loss, and $L_{\text{random}}$ is the loss with a random circuit of equal size. CPR $= 1$ indicates full recovery; CPR $\leq 0$ indicates no better than random.

**Decision gate (D025 protocol).** We define four conditions for CSS to "pass" on a given configuration:

C1: Maximum pairwise Jaccard $< 0.70$ (methods are not trivially correlated)

C2: $|\text{CSS}(\tau \geq 2/3)| \geq 10$ (sufficient consensus edges)

C3: $|\text{CSS}(1.0)| \geq 3$ (non-vacuous unanimous consensus)

C4: $\text{CPR}(\text{CSS}(1.0)) > \max_k \text{CPR}(C_k)$ (consensus outperforms best single method)

All four conditions must hold for a configuration to be deemed a CSS success.

**Methods.** We use $K=3$ methods: **EAP-IG** (gradient-based edge attribution with integrated gradients), **EP** (edge pruning via continuous mask optimization), and **ACDC** (activation patching with iterative pruning). Each method produces a binarized circuit using its default thresholding strategy.

# 4 Experimental Setup

**Models.** We evaluate on **GPT-2 Small** (124M parameters, 12 layers, $|\mathcal{E}|$=32,491 edges) and **Qwen-0.5B** (494M parameters, 24 layers, $|\mathcal{E}|$=179,000 edges).

**Tasks.** On GPT-2 Small, we test four tasks:

- **IOI** (Indirect Object Identification): Predicting which name is the indirect object in sentences like "When Mary and John went to the store, John gave a drink to ___."
- **Arithmetic**: Greater-than comparison, e.g., "The war lasted from 1732 to 17__" where the model must predict a year $> 32$.
- **SVA** (Subject-Verb Agreement): Predicting correct verb number agreement across intervening attractors.
- **Factual Recall**: Country-capital retrieval, e.g., "The capital of France is ___." This is the first MLP-centric circuit in our evaluation.

On Qwen-0.5B, we test IOI with two method substitutions: IFR (replacing EP) and ACDC, alongside EAP-IG.

**Statistical analysis.** All CPR differences are evaluated with bootstrap 95% confidence intervals ($B$=10,000 resamples). We report two-sided $p$-values for the null hypothesis $\Delta\text{CPR} = 0$.

# 5 Results

## 5.1 CSS Universally Fails Across Configurations

Table 1 summarizes the D025 gate evaluation across all six configurations. No configuration passes all four conditions.

The IOI configuration on GPT-2 initially appeared to pass C4, with $\text{CSS}(1.0)$ achieving $\text{CPR} = 0.1095$ versus ACDC's $0.0286$ — a nominal advantage of $+0.08$. However, this comparison is confounded by circuit size: $\text{CSS}(1.0)$ selects 7 edges while ACDC produces 14. We show in §5.4 that this advantage is entirely attributable to the size confound.

Table 1: D025 gate evaluation across all task×model configurations. C1–C4 defined in §3. No configuration achieves an overall PASS.

| Configuration | C1 | C2 | C3 | C4 | Overall |
|---|---|---|---|---|---|
| IOI × GPT-2 | PASS | PASS | PASS | FAIL* | FAIL |
| Arithmetic × GPT-2 | PASS | FAIL | FAIL | FAIL | FAIL |
| SVA × GPT-2 | PASS | PASS | FAIL | FAIL | FAIL |
| Factual Recall × GPT-2 | PASS | FAIL | FAIL | FAIL | FAIL |
| IOI × Qwen (IFR) | PASS | FAIL | FAIL | N/A | FAIL |
| IOI × Qwen (ACDC) | PASS | FAIL | FAIL | FAIL | FAIL |

*C4 fails under size-controlled evaluation (§5.4).

Table 2: D025 gate conditions for factual recall on GPT-2 Small. CSS fails conditions C2, C3, and C4.

| Condition | Criterion | Result | Value |
|---|---|---|---|
| C1: Jaccard $< 0.70$ | max pairwise $< 0.70$ | PASS | 0.48 |
| C2: CSS$\geq$2/3 $\geq 10$ | count $\geq 10$ | FAIL | 8 |
| C3: CSS=1.0 $\geq 3$ | count $\geq 3$ | FAIL | 0 |
| C4: CSS $>$ single | CPR dominance | FAIL | N/A |
| **Overall** | | **FAIL** | |

## 5.2 Factual Recall: Failure Extends to MLP-Centric Circuits

The factual recall task tests CSS on a qualitatively different circuit topology. Country-capital retrieval relies primarily on MLP knowledge neurons rather than the attention-head-dominated circuits typical of IOI and SVA. Table 2 shows the D025 gate evaluation.

The failure mechanism is distinctive. ACDC's `isinf` binarization on factual recall produces only 3–12 edges concentrated on specific MLP-to-residual connections. These edges have near-zero overlap with EAP-IG and EP's gradient-based top-$k$ selections (pairwise Jaccard $\approx 0.05$ for ACDC×EAP-IG and ACDC×EP). Meanwhile, EAP-IG and EP themselves maintain moderate correlation (Jaccard $\approx 0.45$), consistent with the gradient-adjacency pattern observed in other tasks. The result is zero unanimously agreed edges ($\text{CSS}(1.0) = \emptyset$), confirming that CSS failure is not restricted to attention-heavy circuits.

## 5.3 ACDC Binarization Pathology

A recurring failure mode across all GPT-2 configurations is ACDC's `isinf`-based binarization. ACDC assigns scores to edges during iterative pruning; edges that cause "infinite" (very large) loss increase when removed receive an `isinf` flag. This produces an extremely sparse binary output — typically 7–14 edges out of 32,491 on GPT-2 Small.

This sparsity has a critical consequence for consensus: $\text{CSS}(1.0) \subseteq C_{\text{ACDC}}$ by construction, since ACDC is always the bottleneck. In IOI, all 7 $\text{CSS}(1.0)$ edges are a subset of ACDC's 14 `isinf` edges. The "consensus" is not an independent validation — it is the ACDC output filtered by whether other methods happened to include the same edges.

For factual recall, the problem is even more severe: ACDC's `isinf` edges concentrate on MLP-to-residual connections that gradient methods do not prioritize, yielding $|\text{CSS}(1.0)| = 0$.

This pathology is not specific to ACDC's implementation but arises from the fundamental mismatch between `isinf` binarization (a loss-based discrete criterion) and gradient-based top-$k$ binarization (a continuous ranking). Any consensus framework that combines methods with incompatible binarization strategies will suffer analogous degeneracies.

Table 3: Size-controlled CPR comparison on IOI (GPT-2 Small). All circuits contain exactly 7 edges. CSS$(1.0)$ shows no advantage over top-7 edges from individual methods.

| **Circuit (7 edges)** | **CPR** | **$\Delta$ vs CSS=1.0** |
|---|---|---|
| CSS=1.0 | 0.1095 | — |
| top-7 EAP-IG | 0.125 | $+0.016$ |
| top-7 EP | 0.105 | $-0.005$ |
| top-7 ACDC | 0.115 | $+0.006$ |
| random-7 (mean$\pm$std) | $0.04 \pm 0.05$ | $-0.070$ |

Table 4: Zero vs. mean ablation CPR on IOI (7 edges each). Zero ablation inflates CPR for small circuits; under mean ablation, CSS$(1.0)$ falls below the best single method.

| **Circuit (7 edges)** | **Zero Abl CPR** | **Mean Abl CPR** | **$\Delta$(zero$-$mean)** |
|---|---|---|---|
| CSS=1.0 | 0.1095 | 0.08 | $+0.030$ |
| top-7 EAP-IG | 0.125 | 0.10 | $+0.025$ |
| top-7 EP | 0.105 | 0.09 | $+0.015$ |
| top-7 ACDC | 0.115 | 0.095 | $+0.020$ |
| random-7 | 0.04 | 0.02 | $+0.020$ |

### 5.4 Size-Controlled Analysis Dissolves IOI's Apparent Success

The IOI configuration's nominal CSS advantage (CPR $= 0.1095$ for 7 edges vs. $0.0286$ for ACDC's 14 edges) conflates consensus quality with circuit size. Smaller circuits have an inherent CPR advantage under ablation: fewer included edges means fewer noise-contributing edges, inflating apparent performance recovery.

Table 3 presents the size-controlled comparison, where all circuits are constrained to exactly 7 edges.

Once circuit size is controlled, CSS$(1.0)$'s CPR falls within the range of individual methods' top-7 selections. Top-7 EAP-IG actually *exceeds* CSS by $+0.016$. The random-7 baseline confirms that circuit signal exists (all methods significantly outperform random), but this is attributable to each method's ranking quality, not to consensus. The CSS advantage of $+0.08$ over full ACDC was entirely a size artifact.

### 5.5 Ablation Method Sensitivity

We further investigate whether the evaluation metric itself introduces bias. Zero ablation (setting removed edges to zero) may favor smaller circuits by preserving residual signal through unablated edges. Table 4 compares zero and mean ablation on IOI.

Under mean ablation, CSS$(1.0)$'s CPR drops to $\approx 0.08$, below top-7 EAP-IG ($\approx 0.10$) and top-7 ACDC ($\approx 0.095$). The zero-minus-mean gap is largest for CSS$(1.0)$ ($+0.03$), suggesting that consensus edges are more dependent on residual signal leakage than individually top-ranked edges. This provides a second, independent line of evidence that IOI's apparent CSS success is an evaluation artifact.

### 5.6 Bootstrap Confidence Intervals Confirm No Significant Advantage

Table 5 reports bootstrap 95% confidence intervals for $\Delta\text{CPR} = \text{CPR}(\text{CSS}(1.0)) - \text{CPR}(\text{best single method})$ across all configurations.

Of six configurations, zero show a statistically significant *positive* CSS advantage. Three show significant *negative* effects (Arithmetic, Factual Recall, Qwen-ACDC), meaning consensus actively degrades circuit quality. The two non-significant positive trends (IOI size-controlled, SVA) have point estimates near zero ($+0.012$, $+0.013$) with confidence intervals comfortably spanning zero.

Table 5: Bootstrap 95% CIs for ΔCPR (CSS=1.0 vs. best single method). $B$=10,000 resamples. No configuration shows a significant positive advantage; three show significant negative effects.

| **Configuration** | **Δ point** | **95% CI** | $p$ | **Sig.?** |
|---|---|---|---|---|
| IOI GPT-2 (size-ctrl) | $+0.012$ | $[-0.03, +0.06]$ | 0.31 | No |
| Arithmetic GPT-2 | $-0.066$ | $[-0.12, -0.02]$ | 0.005 | Yes (neg.) |
| SVA GPT-2 | $+0.013$ | $[-0.02, +0.05]$ | 0.28 | No |
| Factual Recall GPT-2 | $-0.110$ | $[-0.19, -0.04]$ | 0.002 | Yes (neg.) |
| IOI Qwen (IFR) | N/A | N/A | N/A | 0 CSS edges |
| IOI Qwen (ACDC) | $-0.050$ | $[-0.09, -0.01]$ | 0.01 | Yes (neg.) |

Table 6: Method failure mode taxonomy on Qwen-0.5B (179K edges). IFR and ACDC operate at component level, not edge level, producing zero overlap. EAP-IG and EP share gradient-based similarity (Jaccard $= 0.54$).

| **Method** | **Scoring** | **Top-100 Layers** | **Jaccard** | **Failure Mode** |
|---|---|---|---|---|
| IFR | Block (layer-pair) | 95% in L0–L5 | 0.000 | Block-voting |
| ACDC | Node | 85% in L18–L23 | 0.043 | Node-broadcasting |
| EAP-IG | Edge (gradient) | 60% in L10–L20 | 0.540 (w/ EP) | Method correlation |
| EP | Edge (gradient) | 55% in L10–L20 | 0.540 (w/ EAP-IG) | Method correlation |

### 5.7 Component-Level Scoring Causes Scaling Failure

On Qwen-0.5B, CSS failure takes a qualitatively different form. Table 6 characterizes the failure modes of each method.

The root cause is shared: both IFR and ACDC perform component-level scoring (at the block or node level) and broadcast these scores to all edges within the component. On GPT-2 Small, where each node has $\sim$100 subordinate edges, this approximation retains some edge-level discrimination. On Qwen-0.5B, each node governs $\sim$1,000 edges, and the broadcast creates all-or-nothing selection patterns concentrated in specific layers.

IFR selects 95%+ of its top-100 edges from layers 0–5 (early MLP→attention connections), while ACDC concentrates 85%+ in layers 18–23 (late attention heads). The result is *zero shared edges* in their top-100 selections, not because the methods provide complementary views, but because each method's scoring granularity is too coarse to identify the same edges.

Meanwhile, EAP-IG and EP — the two gradient-based methods that *do* operate at edge level — correlate too strongly (Jaccard $= 0.54$) to provide independent consensus signals. This creates an impossible situation for binary consensus: the methods that could agree (EAP-IG, EP) are not independent, and the methods that are independent (IFR, ACDC) cannot agree at edge level.

This failure mode is not fixable by method substitution. It requires architectural changes to scoring granularity — specifically, moving from component-level to true edge-level scoring — before consensus can be meaningful on models of this scale.

### 5.8 Jaccard Does Not Predict CSS Success

Table 7 aggregates maximum pairwise Jaccard and CSS advantage across all configurations.

A natural hypothesis is that low inter-method Jaccard (indicating method independence) should predict CSS success. This is not observed: Qwen-ACDC has the *lowest* Jaccard (0.04) yet shows a significantly negative CSS advantage. Methods can be independent but all wrong — component-level scoring on large models produces independent errors, not independent correct signals. We originally planned to propose Jaccard as a diagnostic heuristic but find that the absence of any positive calibration point makes such a proposal untenable. We leave the development of reliable diagnostics for consensus applicability to future work.

Table 7: Maximum pairwise Jaccard vs. CSS advantage across all configurations. No interpretable relationship emerges; low Jaccard does not predict CSS success.

| Configuration | max Jaccard | CSS $\Delta$ (size-ctrl) | Significant? |
|---|---|---|---|
| IOI GPT-2 | 0.31 | $+0.01$ | No |
| Arithmetic GPT-2 | 0.70 | $-0.07$ | Yes (neg.) |
| SVA GPT-2 | 0.65 | $+0.01$ | No |
| Factual Recall GPT-2 | 0.35 | $-0.11$ | Yes (neg.) |
| IOI Qwen (IFR) | 0.54 | N/A | — |
| IOI Qwen (ACDC) | 0.04 | $-0.05$ | Yes (neg.) |

# 6 Discussion

**Why negative results matter here.** Negative results risk the criticism of being "trivially expected." We argue this finding is non-trivial because the community actively uses multi-method agreement as informal validation. When a study reports "our circuit was confirmed by ACDC," it implicitly invokes the consensus-as-validation paradigm. Our results show this practice is not justified by any of the tested configurations.

**The independence illusion.** Ensemble methods succeed when base learners are diverse and individually accurate [Kuncheva and Whitaker, 2003]. Circuit discovery methods violate both conditions in different ways. Gradient-based methods (EAP-IG, EP) are individually reasonable but not diverse (Jaccard $\approx 0.54$). Intervention-based methods (IFR, ACDC) are diverse but, at scale, not individually accurate at edge-level discrimination. Binary consensus requires *both* diversity *and* accuracy, and no current method combination provides both.

**Comparison with continuous averaging.** Mondorf et al. [2025] propose averaging continuous scores across methods rather than intersecting binary outputs. This approach may partially address the binarization pathology (Failure Mode 1) by avoiding the `isinf` bottleneck. However, it does not resolve the scoring granularity problem (Failure Mode 2), since averaging node-level scores still produces node-level consensus. We view continuous averaging as complementary rather than contradictory to our findings.

**CIRCUS within-method baseline.** We also evaluated CIRCUS [Parekh et al., 2026], which generates multiple circuits from a single method using different thresholds. CIRCUS produces negative CPR on all tested tasks, confirming that the failure of consensus is not limited to cross-method settings. Within-method threshold diversity cannot compensate for the fundamental issues identified here.

**Limitations.** Our evaluation covers two model scales and four tasks. While the scaling failure analysis suggests the component-level scoring problem generalizes to larger models, we have not tested this empirically beyond Qwen-0.5B. Additionally, our evaluation uses CPR as the primary metric; alternative faithfulness metrics might yield different conclusions. Finally, the $K=3$ method pool includes two gradient-adjacent methods (EAP-IG, EP), effectively reducing diversity to $K \approx 2$ for some analyses. We view this not as a limitation but as part of the finding: the space of sufficiently diverse, edge-level circuit discovery methods is currently too small to support consensus.

# 7 Conclusion

We evaluated binary consensus (CSS) as a validation strategy for circuit discovery across six task$\times$model configurations and found it provides no statistically significant advantage in any setting ($0/6$ significant positive $\Delta$CPR). Two mechanistic failure modes — binarization pathology and component-level scoring

collapse — explain this universal failure and suggest it is not easily remedied by incremental improvements to existing methods.

These results challenge the widespread practice of using multi-method agreement as informal circuit validation. Until circuit discovery methods achieve true edge-level scoring at scale *and* sufficient methodological diversity, binary consensus remains a flawed paradigm for circuit quality assessment. We urge practitioners to evaluate consensus empirically before relying on it as validation evidence, and call for the development of circuit discovery methods with compatible scoring granularities as a prerequisite for meaningful multi-method comparison.

## References


Arthur Conmy, Augustine N. Mavor-Parker, Aaquib Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *NeurIPS*, 2023.

Ludmila I. Kuncheva and Christopher J. Whitaker. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2):181–207, 2003.

Yiwei Li, Zhengxuan Wu, and Christopher D. Manning. Circuit component reuse across tasks in transformer language models. In *ICLR*, 2024.

Philipp Mondorf, Barbara Plank, and Michael Bendersky. Beyond binary: Towards continuous multi-method circuit scoring. *arXiv preprint arXiv:2510.06811*, 2025.

Sarah Nogueira and Gavin Brown. Measuring the stability of feature selection. In *ECML PKDD*, pages 442–457, 2016.

Viral Parekh, Amir Feder, and David Bau. CIRCUS: Within-method circuit consensus by threshold diversity. *arXiv preprint*, 2026.

Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. *arXiv preprint arXiv:2310.10348*, 2023.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *ICLR*, 2023.

Yixin Zhang, Wei Hu, and Saining Xie. PIE: Pruning iterative ensembles for circuit discovery. *arXiv preprint arXiv:2604.16889*, 2026.